

the firm's operations further reduce its free cash flows relative to its competitors. These facts suggest constructing the empirical proxy for productivity using gross profits.

The second difference is in the denominator of the ratios. Where Greenblatt's return on invested capital measure uses "invested capital," Novy-Marx's ratio employs total assets. Like Greenblatt's "invested capital," total assets is independent of the capital structure adopted by management, which makes it comparable to the gross profits measure, which is also independent of the firm's capital structure. Where it differs from Greenblatt's ratio is that it treats cash like any other asset, while Greenblatt deducts net cash to arrive at invested capital. This means that Novy-Marx's ratio penalizes companies carrying lots of cash relative to Greenblatt's measure, which rewards them with a better ratio. While Novy-Marx's penalization of companies holding cash might not make sense on a stand-alone basis, it might be useful in a group of value stocks selected using the enterprise multiple. Recall that one of the problems with the enterprise multiple is that it identifies many small "cash boxes"—companies with large net cash holdings relative to their market capitalization—often because the main business has been sold or the business is a legacy in run-off. While these stocks have very little downside because they are mostly cash, they also have little upside. The gross-profits-to-total-assets measure will penalize these companies for holding cash, thus it should be particularly useful at screening out the cash boxes and identifying undervalued, high-return businesses. The question is, do these slight differences make a significant difference to the returns of undervalued stocks? Figure 10.3 shows the performance of portfolios of stocks constructed from the value portfolios in Figure 10.2, and further divided using Novy-Marx's gross-profits-to-total-assets metric. The "High-Quality Value Stocks" portfolio contains half of the stocks from the value portfolio with the highest ratio of gross profits to total assets, and the "Low-Quality Value Stocks" portfolio contains the half with the lowest ratio.

As before, the All Value Stocks portfolio containing all 114 stocks in the value decile returned 12.50 percent compound, and All Stocks returned 8.36 percent compound. The low-quality value stocks underperformed the whole value portfolio, returning 10.79 percent compound over the full period. The high-quality value stocks outperformed, generating 13.37 percent compound over the full period. While the difference between the compound average performances is small, over the 63-year period under examination the high-quality value portfolio turns \$10,000 into almost \$24 million, while the low-quality value portfolio ends with just \$5.7 million. The high-quality value stocks returned more than four times the capital of the low-quality stocks. Figure 10.3 seems to show that using Novy-Marx's